The model:

$$\begin{cases}\partial_t u = d_1\Delta u + d_2\nabla\cdot(u\nabla u_\tau) + f(u), & x\in\Omega,\ t>0,\\ \dfrac{\partial u}{\partial n} = 0, & x\in\partial\Omega,\ t>0,\\ u(x,0)=\phi(x)\geq 0, & x\in\Omega,\ t\in(-2\tau,0).\end{cases} \tag{0.1}$$

where

$$0<\varepsilon<\tau, \qquad f_u<0, \qquad d_1,\mu_n,\bar u,\tau,\varepsilon>0, \qquad d_2\neq 0.$$

$$u_\tau = \int_{-\tau}^{0} h(s)u(x,t+s)ds, \quad \tau\in\mathbb{R}_+:(0,+\infty). \qquad (-\tau,0), \quad (-\varepsilon,\frac{2}{\tau}), \quad (0,0),$$

and

$$h(s) = \begin{cases}\dfrac{2}{\tau(\tau-\varepsilon)}(s+\tau), & -\tau\leq s\leq -\varepsilon\\ -\dfrac{2}{\tau\varepsilon}s, & -\varepsilon\leq s\leq 0\\ 0, & \text{otherwise}\end{cases} \tag{0.2}$$

The characteristic equation at $\bar u$ is

$$\lambda + d_1\mu_n + d_2\bar u\mu_n\int_{-\tau}^{0} h(s)e^{\lambda s}ds - f_u = 0, \tag{0.3}$$

where

$$\int_{-\tau}^{0} h(s)e^{\lambda s}\,ds = \begin{cases}\dfrac{2(\tau-\varepsilon+\varepsilon e^{-\lambda\tau}-\tau e^{-\lambda\varepsilon})}{\tau\varepsilon(\tau-\varepsilon)\lambda^2}, & \lambda\neq 0,\\ 1, & \lambda=0.\end{cases} \tag{0.4}$$

When $\lambda\neq 0$, we have

$$\lambda + d_1\mu_n - f_u + \frac{2d_2\bar u\mu_n\left(\tau-\varepsilon+\varepsilon e^{-\lambda\tau}-\tau e^{-\lambda\varepsilon}\right)}{\tau\varepsilon(\tau-\varepsilon)\lambda^2} = 0. \tag{0.5}$$

i.e.

$$P_0(\lambda) + P_1 e^{-\lambda\tau} + P_2 e^{-\lambda\varepsilon} = 0. \tag{0.6}$$

where

$$\begin{aligned} P_0(\lambda) &= \lambda^3 + (d_1\mu_n - f_u)\lambda^2 + \frac{2d_2\bar u\mu_n}{\tau\varepsilon},\\ P_1 &= \frac{2d_2\bar u\mu_n}{\tau(\tau-\varepsilon)},\\ P_2 &= -\frac{2d_2\bar u\mu_n}{\varepsilon(\tau-\varepsilon)}, \end{aligned}$$

Let $\lambda = i\omega(\omega > 0)$, we have

$$\begin{aligned} -(d_1\mu_n - f_u)\omega^2 + \frac{2d_2\bar{u}\mu_n}{\tau\varepsilon} + \frac{2d_2\bar{u}\mu_n}{\tau(\tau-\varepsilon)}\cos(\omega\tau) - \frac{2d_2\bar{u}\mu_n}{\varepsilon(\tau-\varepsilon)}\cos(\omega\varepsilon) = 0 \\ -\omega^3 - \frac{2d_2\bar{u}\mu_n}{\tau(\tau-\varepsilon)}\sin(\omega\tau) + \frac{2d_2\bar{u}\mu_n}{\varepsilon(\tau-\varepsilon)}\sin(\omega\varepsilon) = 0 \end{aligned} \tag{0.7}$$

Treating τ as a fixed parameter, we can solve

$$\sin(\omega\varepsilon) = \frac{\varepsilon(\tau-\varepsilon)}{2d_2\bar{u}\mu_n}\omega^3 + \frac{\varepsilon}{\tau}\sin(\omega\tau), \tag{0.8}$$

$$\cos(\omega\varepsilon) = \frac{\tau-\varepsilon}{\tau} + \frac{\varepsilon}{\tau}\cos(\omega\tau) - \frac{\varepsilon(\tau-\varepsilon)}{2d_2\bar{u}\mu_n}(d_1\mu_n - f_u)\omega^2. \tag{0.9}$$

From $\sin^2\omega\varepsilon + \cos^2\omega\varepsilon = 1$, we obtain

$$\begin{aligned} G(\omega,\varepsilon) =& \frac{\varepsilon(\tau-\varepsilon)\tau}{4d_2\bar{u}\mu_n}\omega^6 + \frac{\varepsilon(\tau-\varepsilon)\tau(d_1\mu_n - f_u)^2}{4d_2\bar{u}\mu_n}\omega^4 + \varepsilon\sin(\omega\tau)\omega^3 \\ &- (d_1\mu_n - f_u)\big(\tau - \varepsilon + \varepsilon\cos(\omega\tau)\big)\omega^2 + \frac{2d_2\bar{u}\mu_n}{\tau}\big(\cos(\omega\tau) - 1\big) = 0 \end{aligned} \tag{0.10}$$

Theorem 0.1. *If*

$$d_2\big(d_1\mu_n - f_u + d_2\bar{u}\mu_n\big) > 0,$$

that is,

$$d_2 > 0 \qquad \textit{or} \qquad d_2 < -\frac{d_1\mu_n - f_u}{\bar{u}\mu_n},$$

then, for any fixed $\varepsilon \in (0,\tau)$, the equation $G(\omega,\varepsilon) = 0$ has at least one positive root $\omega > 0$.

Proof. Since $G(\omega,\varepsilon)$ is composed of polynomials, $\sin(\omega\tau)$, and $\cos(\omega\tau)$, it is continuous on $(0,+\infty)$. It therefore suffices to show that $G(\omega,\varepsilon)$ has opposite signs near 0^+ and for sufficiently large ω.

For $\omega > 0$, dividing by ω^2 yields

$$\begin{aligned} \frac{G(\omega,\varepsilon)}{\omega^2} =& \frac{\varepsilon(\tau-\varepsilon)\tau}{4d_2\bar{u}\mu_n}\omega^4 + \frac{\varepsilon(\tau-\varepsilon)\tau(d_1\mu_n - f_u)^2}{4d_2\bar{u}\mu_n}\omega^2 + \varepsilon\omega\sin(\omega\tau) \\ &- (d_1\mu_n - f_u)\big(\tau - \varepsilon + \varepsilon\cos(\omega\tau)\big) + \frac{2d_2\bar{u}\mu_n}{\tau}\frac{\cos(\omega\tau) - 1}{\omega^2}. \end{aligned} \tag{0.11}$$

As $\omega \to 0^+$, the first three terms tend to 0, while

$$-(d_1\mu_n - f_u)\big(\tau - \varepsilon + \varepsilon\cos(\omega\tau)\big) \to -(d_1\mu_n - f_u)\tau.$$

Moreover, letting $x = \omega\tau$ and using $\lim\limits_{x\to 0}\frac{1-\cos x}{x^2} = \frac{1}{2}$, we obtain

$$\lim_{\omega\to 0^+}\frac{2d_2\bar{u}\mu_n}{\tau}\frac{\cos(\omega\tau) - 1}{\omega^2} = 2d_2\bar{u}\mu_n\tau\lim_{x\to 0}\frac{\cos x - 1}{x^2} = -d_2\bar{u}\mu_n\tau.$$

Hence

$$\lim_{\omega\to 0^+}\frac{G(\omega,\varepsilon)}{\omega^2} = -\tau\big(d_1\mu_n - f_u + d_2\bar{u}\mu_n\big).$$

Under the condition

$$d_2\big(d_1\mu_n - f_u + d_2\bar{u}\mu_n\big) > 0,$$

the quantity $d_1\mu_n - f_u + d_2\bar{u}\mu_n$ has the same sign as d_2, so

$$-\tau\big(d_1\mu_n - f_u + d_2\bar{u}\mu_n\big)$$

has the opposite sign of d_2. Since $\omega^2 > 0$, it follows that $G(\omega,\varepsilon)$ has the opposite sign of d_2 for all sufficiently small $\omega > 0$.

Since $|\sin(\omega\tau)| \le 1$ and $|\cos(\omega\tau)| \le 1$, letting $\omega \to +\infty$ gives

$$\lim_{\omega\to+\infty}\frac{G(\omega,\varepsilon)}{\omega^6} = \frac{\varepsilon(\tau-\varepsilon)\tau}{4d_2\bar{u}\mu_n}.$$

Therefore, $G(\omega,\varepsilon)$ has the same sign as d_2 for all sufficiently large ω.

Thus, $G(\omega,\varepsilon)$ is continuous on $(0,+\infty)$, has the opposite sign of d_2 near 0^+, and has the same sign as d_2 for sufficiently large ω. By the intermediate value theorem, for any fixed $\varepsilon \in (0,\tau)$, there exists some $\omega > 0$ such that $G(\omega,\varepsilon) = 0$. This completes the proof. $\square$

Let $\omega_n(\varepsilon)$ be a chosen positive simple root of $G(\omega,\varepsilon) = 0$. Then, there exists a unique $\theta_n(\varepsilon) \in [0, 2\pi)$ such that

$$\cos\theta_n(\varepsilon) = \cos(\omega_n(\varepsilon)\varepsilon), \qquad \sin\theta_n(\varepsilon) = \sin(\omega_n(\varepsilon)\varepsilon).$$

Equivalently,

$$\theta_n(\varepsilon) = \arg\big(\cos(\omega_n(\varepsilon)\varepsilon) + i\sin(\omega_n(\varepsilon)\varepsilon)\big).$$

It can also be written in the following piecewise form:

$$\theta_n(\varepsilon) = \begin{cases} \arctan\frac{\sin(\omega_n\varepsilon)}{\cos(\omega_n\varepsilon)}, & if\ \sin(\omega_n\varepsilon) > 0, \cos(\omega_n\varepsilon) > 0, \\ \frac{\pi}{2}, & if\ \sin(\omega_n\varepsilon) = 1, \cos(\omega_n\varepsilon) = 0, \\ \pi + \arctan\frac{\sin(\omega_n\varepsilon)}{\cos(\omega_n\varepsilon)}, & if\ \cos(\omega_n\varepsilon) < 0, \\ \frac{3\pi}{2}, & if\ \sin(\omega_n\varepsilon) = -1, \cos(\omega_n\varepsilon) = 0, \\ 2\pi + \arctan\frac{\sin(\omega_n\varepsilon)}{\cos(\omega_n\varepsilon)}, & if\ \sin(\omega_n\varepsilon) < 0, \cos(\omega_n\varepsilon) > 0, \end{cases} \tag{0.12}$$

Hence,

$$\omega_n(\varepsilon)\varepsilon = \theta_n(\varepsilon) + 2k\pi, \qquad k = 0, 1, 2, \dots.$$

Define

$$S_{n,k}(\varepsilon) := \varepsilon - \frac{\theta_n(\varepsilon) + 2k\pi}{\omega_n(\varepsilon)}.$$

Then the critical values $\varepsilon = \varepsilon_{n,k}$ at which the characteristic equation admits a purely imaginary root can be determined by solving

$$S_{n,k}(\varepsilon) = 0.$$